Generative AI Healthcare Agents

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Data Collection

Key Data Sources

MIMIC-III, IV (PhysioNet)

- 380K+ ICU patients, 2M+ clinical notes
- 6 CSV files: Diagnosis, EDStays, MedRecon, Pyxis, Triage, Vitalsign
- Rows: 7,195,726
- Includes: Vitals, diagnoses, meds, triage scores, demographics
- URL: physionet.org

iCliniq (Hugging Face)

- Online telemedicine consultations
- 39,540 records
- Fields: Title, Question, Answer

MedQuAD (U.S. NLM)

- Expert-verified medical Q&A
- 16,412 records
- Fields: Question, Answer, Source, Focus_area

Storage:** All datasets uploaded to **Google Cloud Platform

Data Ingestion

- **Process** - Uploaded CSV files to GCP Cloud Storage bucket gs://data298a.
- **Security** - Configured IAM roles to restrict access to project team and pipeline services.
- **Features** - Enabled versioning to track data history, Logged metadata (e.g., upload timestamps) for traceability.

Cloud Composer Setup -

- Environment: healthcare-composer (3 nodes, 2 vCPUs, 4 GB memory each).
- Installed PyPI packages: pandas, google-cloud-storage, apache-airflow[gcp].
- Auto-generated GKE cluster and gs://healthcare-composer-bucket/dags.

Centralized storage with ~7M rows, ready for transformation

Data Preprocessing

- **Cleaning:** Removed null values, special characters, excessive whitespace
- **Normalization:** Lowercase text, standardized formats
- **Stop Word Removal:** Removed common words (e.g., *the, is, at*) to reduce noise
- **Stemming & Lemmatization:** Reduced words to their root/base form for semantic consistency
- **Field Merging:** Combined *title + question* (iCliniq), merged multi-source tables (MIMIC)
- **ICD Code Mapping:** Mapped ICD-9/10 codes to clinical descriptions for interpretability
- **Pre-tokenization & Embedding:** Vectorized text fields using BioBERT
- **LLM-Ready Outputs:** Final JSONL formatted for fine-tuning and Retrieval-Augmented Generation (RAG)

Data Transformation

- **Schema Flattening:** Merged MIMIC tables (edstays, diagnosis, vitals, meds, triage) into a single patient-stay record.
- **Feature Engineering:**
  - Extracted ICD categories & chapters from raw ICD-9/10 codes.
  - Integrated diagnostic labels with clinical descriptions.
  - Computed aggregated vitals (mean heart rate, BP, etc.) per patient stay.
- **Field Alignment:** Standardized columns (instruction, input, output) across MedQuAD, iCliniq, and MIMIC QA datasets for consistency.
- **Semantic Enrichment:** Augmented records with ICD descriptions, chief complaints, medications, and demographics.
- **Format Conversion:** Exported into **JSONL** files for LLM training and RAG pipelines.
- **Embedding Preparation:** Generated text embeddings (PubMed-BERT/BioBERT) stored in Pinecone for retrieval. **PubMed (API access)** → Real-time biomedical literature retrieval.

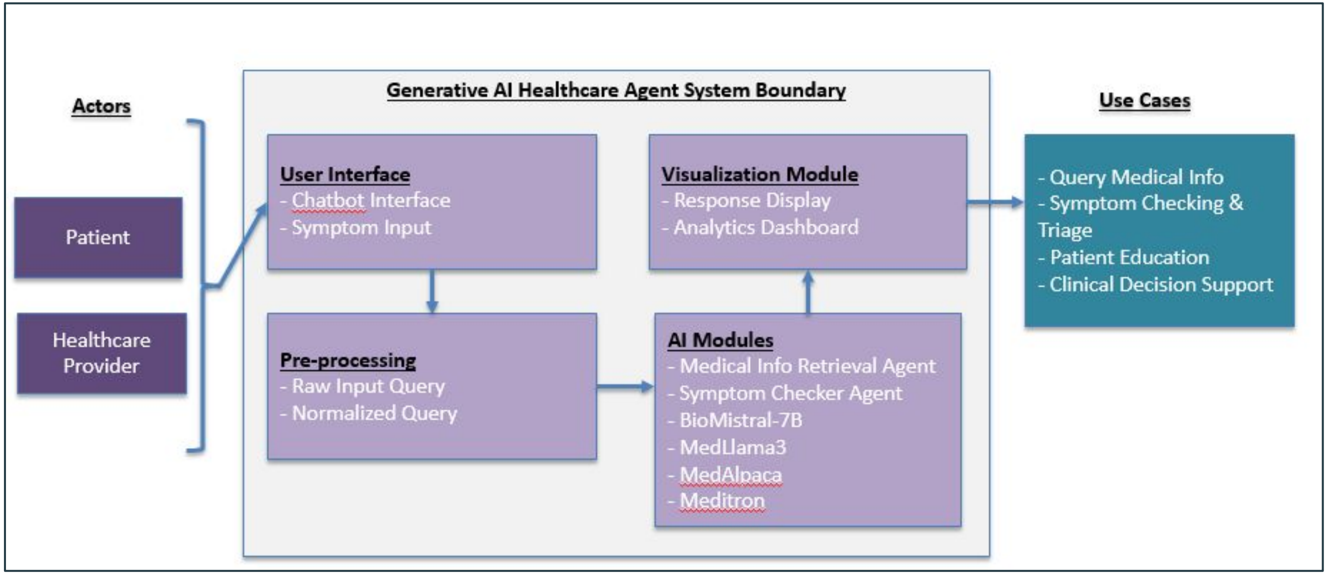
Data Preparation

- **Unified Dataset Creation:** Integrated QA datasets (MIMIC-III, IV, MedQuAD, iCliniq) into a single corpus.
- **Training Format Conversion:** Exported into **JSONL format** with fields:
  - *context* ⟶ background (e.g., title, clinical notes)
  - *input* ⟶ user query/question
  - *output* ⟶ ground-truth answer
- **Data Splitting:** Applied stratified split:
  - 80% ⟶ Training (fine-tuning LLMs)
  - 10% ⟶ Validation (hyperparameter tuning, early stopping)
  - 10% ⟶ Test (final unbiased evaluation)
- **Balance & Consistency Checks:** Verified schema alignment, ensured no data leakage, checked distribution of diagnoses, query lengths, and response types across splits.
- **Hosting & Scalability:** Stored datasets in **Google BigQuery** for large-scale queries and integration with GCP pipelines.

System Requirements Analysis

System Boundary: Clinical datasets, PubMed APIs, AI retrieval & reasoning modules, visualization dashboards.
Actors: Patients, Healthcare providers, Administrators
Use Cases: Medical info retrieval, symptom checking, patient education, clinical decision support.
Capabilities:

- Fine-tuned LLMs (MedLlama3, BioMistral-7B, Med-PaLM, Meditron).
- RAG with Pinecone embeddings.
- Analytics: BLEU, ROUGE, FCS, MCR.
- Real-time PubMed retrieval.
- Visualization of performance metrics and patient interactions.



System Design

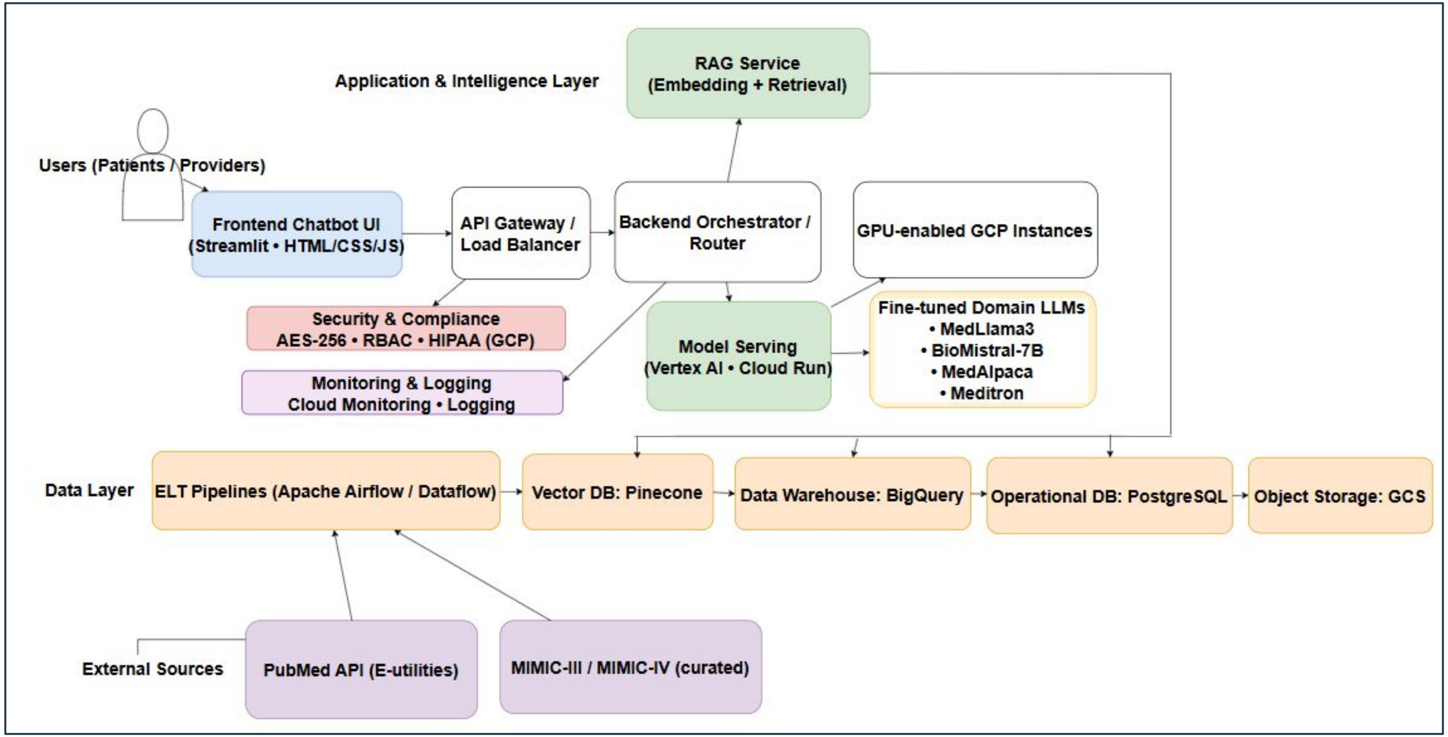
Architecture:

- Front-End: Streamlit chatbot
- Back-End: GCP pipelines with Airflow, Dataflow, BigQuery, Pinecone.
- Deployment: Vertex AI + Cloud Run.
- Security: RBAC.

Platforms: PyTorch, TensorFlow, Hugging Face, Pinecone, PostgreSQL, Tableau.

Data Management: Centralized warehouse in BigQuery + Pinecone for embeddings; secured with HIPAA compliance.

User Interface & Visualization: Chatbot + Tableau dashboards



Model Selection

Selected open-source LLMs optimized for chronic disease management, ensuring accuracy, scalability, and patient engagement.

Selected Models:

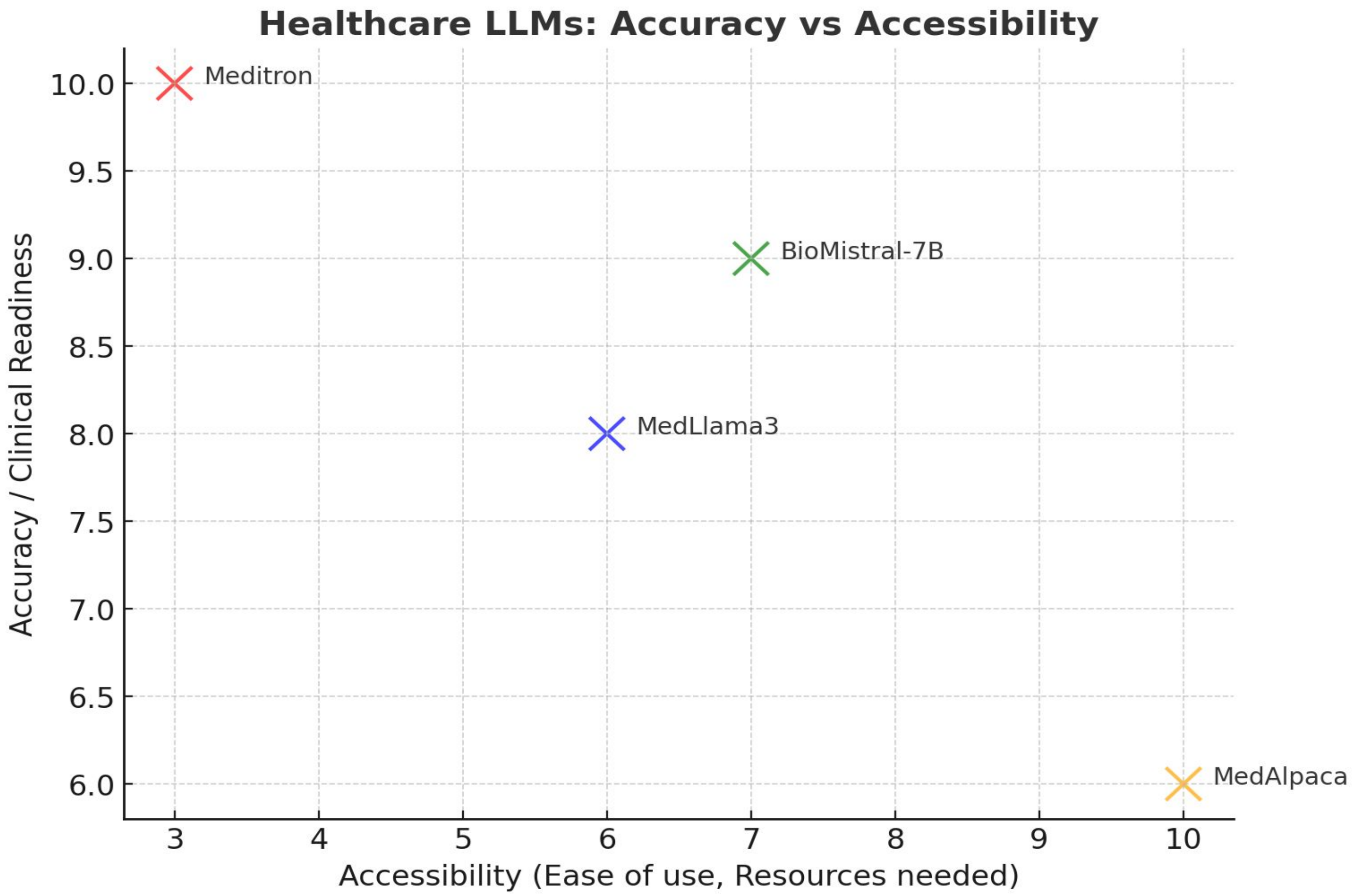
- **MedLlama3:** General-purpose Q&A and patient education
- **BioMistral-7B:** Biomedical knowledge retrieval and clinical research
- **MedAlpaca:** Lightweight generative medical text and Q&A model,
- **Meditron:** Advanced medical reasoning and differential diagnosis

We have currently fine-tuned the BioMistral-7B and MedAlpaca models on healthcare-specific Q/A datasets, to enhance its performance in biomedical knowledge retrieval and clinical research queries.

Model Supports

- **Platform**: Google Cloud Platform (GCP).
- **Front-End**: Streamlit + HTML, CSS, JavaScript chatbot interface.
- **Back-End**: ELT pipeline (Apache Airflow, Dataflow), PostgreSQL + BigQuery for storage.
- **Vector DB**: Pinecone for embeddings.
- **Deployment**: Vertex AI (fine-tuning) + Cloud Run (scalable serving).
- **Security**: HIPAA-compliant RBAC.

Model	Target Problems	Key Features	Approach /	Strengths	Limitations	Data Size / Complexity	Data Types
MedLlama3	General-purpose healthcare Q&A, patient education, preliminary health advice	Built on LLaMA-2; versatile in Q&A; handles complex medical terminology via self-attention	Transformer-based deep learning with fine-tuning on healthcare datasets	High accuracy and versatility; ideal for diverse medical queries & patient-facing interactions	Computationally intensive; may require high resources for real-time deployment in low-resource settings	High data size, complex queries	General medical queries, patient education materials, clinical notes
BioMistral-7B	Biomedical knowledge retrieval, clinical research Q&A, specialized medical queries	Pre-trained on PubMed; domain-specific language; supports multilingual queries; model-merging (SLERP, DARE)	NLP & deep learning with domain-adaptive pre-training	Exceptional in biomedical retrieval & terminology; excels in research-oriented queries	Limited real-time support due to complexity; less effective for general health queries	Moderate–large, complex biomedical terms	Biomedical research data (PubMed), clinical guidelines, research papers
Meditron	Advanced reasoning, differential diagnosis, clinical decision support	Built on LLaMA-2/3.1; pre-trained on GAP-Replay (48.1B tokens); supports multimodal tasks (Meditron-V)	Transformer-based with domain-adaptive pre-training & in-context learning	High accuracy (77.6% on MedQA); supports complex reasoning; scalable (7B–70B); humanitarian applications	Research-only; requires clinical validation; resource-heavy (esp. 70B variant)	Very high data size, complex diagnostic queries	Clinical guidelines, PubMed abstracts, full-text medical papers, structured clinical datasets
MedAlpaca	Generative medical text, patient education, research/educational Q&A	Instruction-tuned on LLaMA using Alpaca method; trained on PubMed abstracts, Wikipedia medical pages, clinical guidelines	Autoregressive transformer fine-tuned with instruction datasets	Lightweight, fully open-source; easy to fine-tune for research/education; accessible baseline model	Lower accuracy than domain-specific models ; not clinically validated	Small–moderate data size; moderate query complexity	PubMed abstracts, Wikipedia medical text, guideline-based QA


Healthcare LLMs: Accuracy vs Accessibility
Meditron
BioMistral-7B
MedLlama3
MedAlpaca
Accuracy / Clinical Readiness
Accessibility (Ease of use, Resources needed)
10.0
9.5
9.0
8.5
8.0
7.5
7.0
6.5
6.0
3
4
5
6
7
8
9
10

Comparison of BioMistral Baseline and fine tuned

Metric	Baseline	Fine-Tuned
BERTScore Precision	0.8	0.8
BERTScore Recall	0.86	0.89
BERTScore F1	0.83	0.84
FCS (Factual Consistency Score)	0.3859	0.53
Hybrid BERT-BLEU	0.58	0.59
LLM Judge Score	0.52	0.84
GEval Score	0.44	0.75
METEOR	0.16	0.24
ROUGE-L	0.08	0.08
Entity F1	0.2	0.1
MCR (Medical Concept Recall)	0.2	0.2
BLEU	0.01	0.02

Comparison of BioMistral and MedAlpaca

Metric	BioMistral	MedAlpaca
BERTScore Precision	0.8	0.5
BERTScore Recall	0.89	0.52
BERTScore F1	0.84	0.5
FCS (Factual Consistency Score)	0.53	0.26
Hybrid BERT-BLEU	0.59	0.31
LLM Judge Score	0.84	0.42
GEval Score	0.75	0.31
METEOR	0.24	0.1
ROUGE-L	0.08	0.05
Entity F1	0.1	0.1
MCR (Medical Concept Recall)	0.2	0.1
BLEU	0.02	0.01

Thank You